GLM-5 W8A8 A2部署调优实践

关键词：GLM-5、vLLM-Ascend、Atlas 800 A2、性能调优

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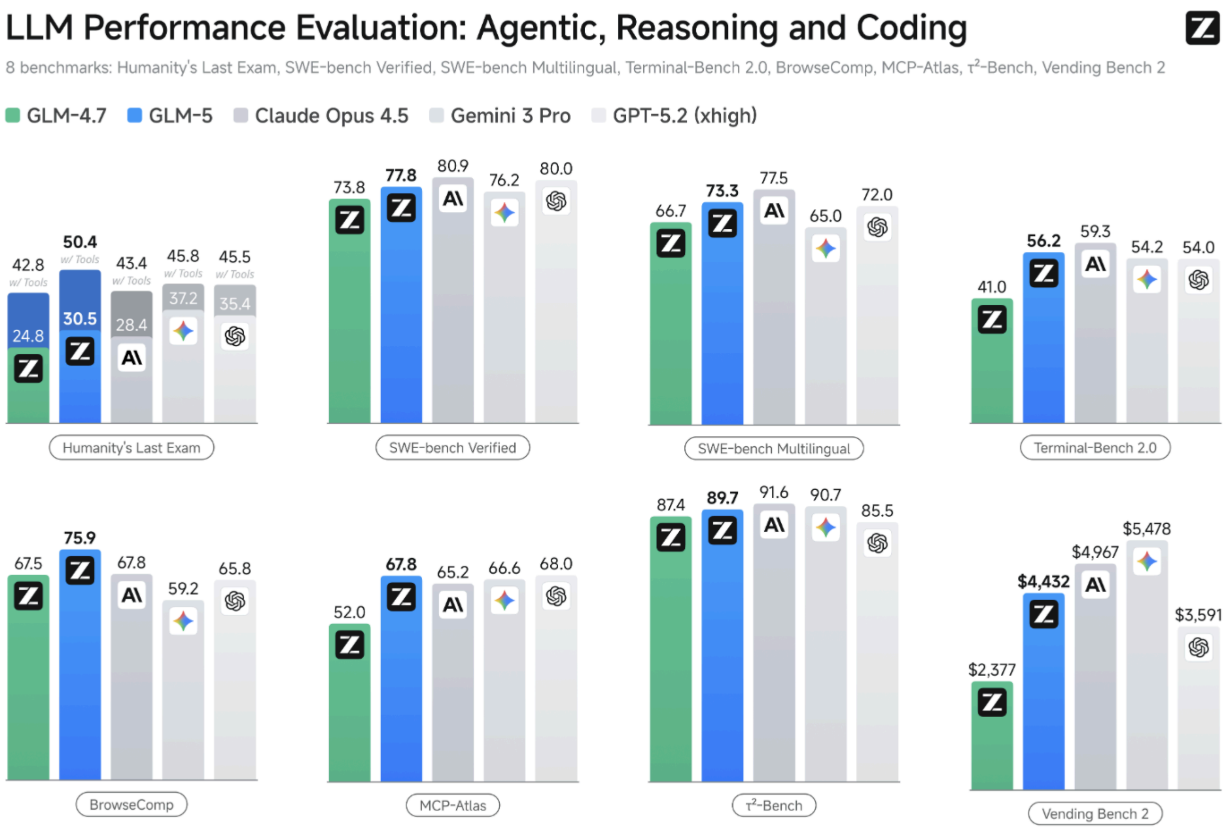
1. 背景描述

1.1 模型描述

[GLM-5](#) 是智谱新一代的旗舰基座模型，面向 Agentic Engineering 打造，能够在复杂系统工程与长程 Agent 任务中提供可靠生产力。在 Coding 与 Agent 能力上，GLM-5 取得开源 SOTA 表现，在真实编程场景的使用体感逼近 Claude Opus 4.5，擅长复杂系统工程与长程 Agent 任务，是通用 Agent 助手的理想基座。



1.2 环境信息

组件	版本	备注
服务器硬件	Atlas 800 A2	8机PD分离、4机混部
vLLM-Ascend	quay.io/ascend/vllm-ascend:main	合入PR [7139](#)
HDK	25.2.0	
CANN	8.2.RC1	

组件	版本	备注
GLM-5权重	W8A8量化	https://modelscope.cn/models/umiiiiii/GLM-W8A8/files

2. 叠加特性优化

优化特性	使能方法
w8a8模型量化	https://modelscope.cn/models/umiiiiii/GLM-W8A8/files
FLASHCOMM1算子接入	export VLLM_ASCEND_ENABLE_FLASHCOMM1=1
异步调度	vllm启动命令中--async-scheduling
MLAPO算子接入	https://github.com/vllm-project/vllm-ascend/pull/6902 适配接入
mul_add融合算子使能	https://github.com/vllm-project/vllm-ascend/pull/5518https://github.com/vllm-project/vllm-ascend/pull/6928 适配接入 --additional-config中加"fuse_muls_add": true
PD分离	'{ 【1P1D】 P：DP4、TP8，D：DP8、TP4
共享专家多流	--additional-config中加 "recompute_scheduler_enable":true "multistream_overlap_shared_expert": true
MTP接受率提升	"multistream_overlap_shared_expert":true, https://github.com/yydyzr/vllm/pull/2" --additional-config中加rot_path": "xxx/rot.safetensors" --speculative-config '{"num_speculative_tokens": 1, "method":"deepseek_mtp"}'
MTP-DP入图	"fuse_qknorm_rope": false, https://github.com/vllm-project/vllm-ascend/pull/6948
流水优化	"fuse_muls_add": true, export TASK_QUEUE_ENABLE=1
通信算法 - "AIV"	--additional-config中加"enable_npugraph_ex": true export HCCL_OP_EXPANSION_MODE="AIV"
FULL_DECODE_ONLY	【仅D节点】--compilation-config '{"cudagraph_mode":"FULL_DECODE_ONLY", "cudagraph_capture_sizes":[4, 8, 12, 16, 20, 24, 28, 32]}'

3. 部署指导

3.1 创建容器

创建docker.sh启动脚本

```
export IMAGE=a480062695ec
```

```
docker run  --privileged \
    --name glm-5 \
    --shm-size=500g \
    --net=host \
    --device /dev/davinci0 \
    --device /dev/davinci1 \
    --device /dev/davinci2 \
    --device /dev/davinci3 \
    --device /dev/davinci4 \
    --device /dev/davinci5 \
    --device /dev/davinci6 \
    --device /dev/davinci7 \
    --device /dev/davinci_manager \
    --device /dev/devmm_svm \
    --device /dev/hisi_hdc \
    -v /usr/share/zoneinfo/Asia/Shanghai:/etc/localtime \
    -v /usr/local/dcmi:/usr/local/dcmi \
    -v /usr/local/Ascend/driver/tools/hccn_tool:/usr/local/Ascend/driver/tools/hccn_tool \
    -v /usr/local/bin/npu-smi:/usr/local/bin/npu-smi \
    -v /usr/local/Ascend/driver/lib64/:/usr/local/Ascend/driver/lib64/ \
    -v /usr/local/Ascend/driver/version.info:/usr/local/Ascend/driver/version.info \
    -v /etc/ascend_install.info:/etc/ascend_install.info \
    -v /root/.cache:/root/.cache \
    -v /etc/hccn.conf:/etc/hccn.conf \
    -v /mnt/:/mnt/ \
    -v /opt/:/opt/ \
    -v /home:/home \
    -itd $IMAGE bash
```

执行

```
# 运行容器
bash docker.sh

# 进入容器
docker exec -it glm-5 bash
```

3.2 PD分离部署（8机、1P1D）

并行策略

- P实例：DP4，TP8
- D实例：DP8，TP4

组网

节点	IP	需要文件
P节点（主）	71.10.29.138	launch_online_dp.py、run_dp_template.sh、server.sh、proxy.sh、load_balance_proxy_server_example.py
P节点（从）	71.10.29.141	launch_online_dp.py、run_dp_template.sh、server.sh

节点	IP	需要文件
P节点（从）	71.10.29.125	launch_online_dp.py、run_dp_template.sh、server.sh
P节点（从）	71.10.29.128	launch_online_dp.py、run_dp_template.sh、server.sh
D节点（主）	71.10.29.124	launch_online_dp.py、run_dp_template.sh、server.sh
D节点（从）	71.10.29.123	launch_online_dp.py、run_dp_template.sh、server.sh
D节点（从）	71.10.29.139	launch_online_dp.py、run_dp_template.sh、server.sh
D节点（从）	71.10.29.142	launch_online_dp.py、run_dp_template.sh、server.sh

脚本

launch_online_dp.py： 每个节点都要有，无需修改

run_dp_template.sh：每个节点根据实际情况修改

dp_load_balance_proxy_server.py：仅P主节点需要

详细说明见：https://github.com/vllm-project/vllm-ascend/blob/main/examples/external_online_dp/README.md

3.2.1 P节点

run_dp_template.sh模板，请按实际情况修改`nic_name`、`local_ip`、`权重路径`、`rot_path`、`torch_profiler_dir`

```
rm -rf ~/ascend

export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/local/lib
export PYTHONPATH=/workspace/vllm-ascend_3_5/vllm-ascend:${PYTHONPATH}

export VLLM_ASCEND_ENABLE_MLAPO=1
export VLLM_ASCEND_ENABLE_NZ=1

export HCCL_OP_EXPANSION_MODE="AIV"

nic_name="enp67s0f0np0"
local_ip=71.10.29.138

export HCCL_IF_IP=$local_ip
export GLOO_SOCKET_IFNAME=$nic_name
export TP_SOCKET_IFNAME=$nic_name
export HCCL_SOCKET_IFNAME=$nic_name

#Mooncake
export OMP_PROC_BIND=false
export OMP_NUM_THREADS=10
export ASCEND_CONNECT_TIMEOUT=300000
export ASCEND_TRANSFER_TIMEOUT=300000
export ASCEND_BUFFER_POOL=4:8
```

```
export LD_LIBRARY_PATH=/usr/local/Ascend/ascend-toolkit/latest/python/site-
packages/mooncake:$LD_LIBRARY_PATH

export VLLM_USE_V1=1
export HCCL_BUFFSIZE=200
export PYTORCH_NPU_ALLOC_CONF=expandable_segments:True

export VLLM_ASCEND_BALANCE_SCHEDULING=1

# optim
export TASK_QUEUE_ENABLE=1
export CPU_AFFINITY_CONF=1

export VLLM_ASCEND_ENABLE_FLASHCOMM1=1
export VLLM_ASCEND_ENABLE_FUSED_MC2=1
export ASCEND_AGGREGATE_ENABLE=1
export ASCEND_TRANSPORT_PRINT=1
export ACL_OP_INIT_MODE=1
export VLLM_NIXL_ABORT_REQUEST_TIMEOUT=300000

export ASCEND_RT_VISIBLE_DEVICES=$1

vllm serve /opt/data/verification/models/GLM-W8A8 \
    --host 0.0.0.0 \
    --port $2 \
    --data-parallel-size $3 \
    --data-parallel-rank $4 \
    --data-parallel-address $5 \
    --data-parallel-rpc-port $6 \
    --tensor-parallel-size $7 \
    --enable-expert-parallel \
    --enable-chunked-prefill \
    --seed 1024 \
    --served-model-name glm5 \
    --max-model-len 133120 \
    --max-num-batched-tokens 4096 \
    --trust-remote-code \
    --max-num-seqs 32 \
    --gpu-memory-utilization 0.95 \
    --quantization ascend \
    --async-scheduling \
    --enforce-eager \
    --enable-auto-tool-choice \
    --tool-call-parser glm47 \
    --reasoning-parser glm45 \
    --kv-transfer-config \
    '{
        "kv_connector": "MooncakeConnectorV1",
        "kv_role": "kv_producer",
        "kv_port": "30000",
        "engine_id": "0",
        "kv_connector_module_path": "vllm_ascend.distributed.mooncake_connector",
        "kv_connector_extra_config": {
```

```
            "use_ascend_direct": true,
            "prefill": {
                "dp_size": 4,
                "tp_size": 8
            },
            "decode": {
                "dp_size": 8,
                "tp_size": 4
            }
        }
    }' \
    --additional-config \
    '{
        "recompute_scheduler_enable":true,
        "multistream_overlap_shared_expert":true,
        "fuse_qknorm_rope": false,
        "fuse_muls_add": true,
        "enable_npugraph_ex": true,
        "rot_path": "/home/glm5/scripts/rot.safetensors"
    }' \
    --speculative-config '{"num_speculative_tokens": 1, "method":"deepseek_mtp"}' \
    --profiler-config \
    '{
        "profiler": "torch",
        "torch_profiler_dir": "/home/glm5/profiling",
        "torch_profiler_with_stack": false
    }' \
    2>&1 | tee glm.log
```

server.sh： P实例 DP4、TP8

```
# 71.10.29.138 P主节点
python launch_online_dp.py --dp-size 4 --tp-size 8 --dp-size-local 1 --dp-rank-start 0 --dp-address 71.10.29.138 --dp-rpc-port 10521 --vllm-start-port 6600

# 71.10.29.141 P从节点
python launch_online_dp.py --dp-size 4 --tp-size 8 --dp-size-local 1 --dp-rank-start 1 --dp-address 71.10.29.138 --dp-rpc-port 10521 --vllm-start-port 6600

# 71.10.29.125 P从节点
python launch_online_dp.py --dp-size 4 --tp-size 8 --dp-size-local 1 --dp-rank-start 2 --dp-address 71.10.29.138 --dp-rpc-port 10521 --vllm-start-port 6600

# 71.10.29.128 P从节点
python launch_online_dp.py --dp-size 4 --tp-size 8 --dp-size-local 1 --dp-rank-start 3 --dp-address 71.10.29.138 --dp-rpc-port 10521 --vllm-start-port 6600
```

proxy.sh

- 只存在于P主节点
- 在P/D实例服务启动成功后执行，`bash proxy.sh > proxy.log &`
- 根据实际情况修改组网IP

```
unset http_proxy
unset https_proxy

python load_balance_proxy_server_example.py \
    --port 8000 \
    --host 0.0.0.0 \
    --prefiller-hosts \
       71.10.29.138 \
       71.10.29.141 \
       71.10.29.128 \
       71.10.29.125 \
    --prefiller-ports \
       6600 \
       6600 \
       6600 \
       6600 \
    --decoder-hosts \
       71.10.29.124 \
       71.10.29.124 \
       71.10.29.142 \
       71.10.29.142 \
       71.10.29.139 \
       71.10.29.139 \
       71.10.29.123 \
       71.10.29.123 \
    --decoder-ports \
       6600 6601 \
       6600 6601 \
       6600 6601 \
       6600 6601
```

3.2.2 D节点

run_dp_template.sh模板，请按实际情况修改`nic_name`、`local_ip`、`权重路径`、`rot_path`、`torch_profiler_dir`

```
rm -rf ~/ascend

export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/local/lib
export PYTHONPATH=/workspace/vllm-ascend_3_5/vllm-ascend:${PYTHONPATH}

export VLLM_ASCEND_ENABLE_MLAPO=1
export VLLM_ASCEND_ENABLE_NZ=1

export HCCL_OP_EXPANSION_MODE="AIV"

nic_name="enp67s0f0np0"
local_ip=71.10.29.124

export HCCL_IF_IP=$local_ip
export GLOO_SOCKET_IFNAME=$nic_name
export TP_SOCKET_IFNAME=$nic_name
```

```
export HCCL_SOCKET_IFNAME=$nic_name

#Mooncake
export OMP_PROC_BIND=false
export OMP_NUM_THREADS=10
export ASCEND_CONNECT_TIMEOUT=300000
export ASCEND_TRANSFER_TIMEOUT=300000
export ASCEND_BUFFER_POOL=4:8
export LD_LIBRARY_PATH=/usr/local/Ascend/ascend-toolkit/latest/python/site-
packages/mooncake:$LD_LIBRARY_PAT
export VLLM_USE_V1=1
export HCCL_BUFFSIZE=200

export PYTORCH_NPU_ALLOC_CONF=expandable_segments:True
export VLLM_ASCEND_BALANCE_SCHEDULING=1

# optim
export TASK_QUEUE_ENABLE=1
export CPU_AFFINITY_CONF=1

export VLLM_ASCEND_ENABLE_FUSED_MC2=1
export ASCEND_AGGREGATE_ENABLE=1
export ASCEND_TRANSPORT_PRINT=1
export ACL_OP_INIT_MODE=1
export VLLM_NIXL_ABORT_REQUEST_TIMEOUT=300000


export ASCEND_RT_VISIBLE_DEVICES=$1



vllm serve /opt/data/verification/models/GLM-W8A8 \
    --host 0.0.0.0 \
    --port $2 \
    --data-parallel-size $3 \
    --data-parallel-rank $4 \
    --data-parallel-address $5 \
    --data-parallel-rpc-port $6 \
    --tensor-parallel-size $7 \
    --enable-expert-parallel \
    --seed 1024 \
    --served-model-name glm5 \
    --max-model-len 133120 \
    --max-num-batched-tokens 32 \
    --trust-remote-code \
    --max-num-seqs 32 \
    --gpu-memory-utilization 0.95 \
    --async-scheduling \
    --quantization ascend \
    --enable-auto-tool-choice \
    --tool-call-parser glm47 \
    --reasoning-parser glm45 \
    --kv-transfer-config \
    '{
```

```
        "kv_connector": "MooncakeConnectorV1",
        "kv_role": "kv_consumer",
        "kv_port": "30100",
        "engine_id": "1",
        "kv_connector_module_path": "vllm_ascend.distributed.mooncake_connector",
        "kv_connector_extra_config": {
                "use_ascend_direct": true,
                "prefill": {
                        "dp_size": 4,
                        "tp_size": 8
                },
                "decode": {
                        "dp_size": 8,
                        "tp_size": 4
                }
        }
    }' \
    --compilation-config \
    '{
        "cudagraph_capture_sizes": [1,4,8,12,16,20,24,28,32,36,40,48,56,64,80,96],
        "cudagraph_mode": "FULL_DECODE_ONLY"
     }' \
    --additional-config \
    '{
        "recompute_scheduler_enable":true,
        "multistream_overlap_shared_expert":true,
        "fuse_qknorm_rope": false,
        "fuse_muls_add": true,
        "enable_npugraph_ex": true,
        "rot_path": "/home/glm5/scripts/rot.safetensors"
    }' \
    --speculative-config '{"num_speculative_tokens": 3,  "method":"deepseek_mtp"}' \
    --profiler-config \
    '{
        "profiler": "torch",
        "torch_profiler_dir": "/home/glm5/profiling",
        "torch_profiler_with_stack": false
     }' \
    2>&1 | tee glm.log
```

server.sh： D实例 DP8、TP4

```
# 71.10.29.124 D主节点
python launch_online_dp.py --dp-size 8 --tp-size 4 --dp-size-local 2 --dp-rank-start 0 --dp-address 71.10.29.124 --dp-rpc-port 10521 --vllm-start-port 6600

# 71.10.29.123 D从节点
python launch_online_dp.py --dp-size 8 --tp-size 4 --dp-size-local 2 --dp-rank-start 2 --dp-address 71.10.29.124 --dp-rpc-port 10521 --vllm-start-port 6600

# 71.10.29.139 D从节点
python launch_online_dp.py --dp-size 8 --tp-size 4 --dp-size-local 2 --dp-rank-start 4 --dp-address 71.10.29.124 --dp-rpc-port 10521 --vllm-start-port 6600

# 71.10.29.142 D从节点
python launch_online_dp.py --dp-size 8 --tp-size 4 --dp-size-local 2 --dp-rank-start 6 --dp-address 71.10.29.124 --dp-rpc-port 10521 --vllm-start-port 6600
```

4. 性能基线

4.1 P-DP4-TP8_D-DP8-TP4_Prefix cache 0%-MTP3（无优化）

输入	输出	并发	Mean TTFT (ms)	Median TTFT (ms)	P90 TTFT (ms)	P99 TTFT (ms)	Mean TPOT (ms)	Median TPOT (ms)	P90 TPOT (ms)	P99 TPOT (ms)	Mean ITL (ms)	Median ITL (ms)	P90 ITL (ms)	P99 ITL (ms)
3.5K	1.5K	10	2249.30	1948.79	3704.12	4171.17	49.28	46.39	64.62	79.00	79.31	75.47	99.99	193.04
32K	1K	10	13858.68	11791.30	19927.97	27614.58	76.04	76.98	79.00	80.33	80.07	77.24	106.66	194.77
32K	2K	10	13381.57	11622.34	19549.19	26983.33	75.71	76.09	77.58	77.84	79.14	77.18	103.95	173.72
64K	1K	10	26439.01	22437.97	38042.62	55339.73	79.57	80.74	83.13	85.40	82.89	80.62	105.44	184.02
64K	2K	10	25695.28	22302.24	39395.78	56534.29	77.60	79.96	81.40	82.17	82.10	80.56	108.12	168.63

4.2 P-DP4-TP8_D-DP8-TP4_Prefix cache 90%-MTP3（无优化）

输入	输出	并发	Mean TTFT (ms)	Median TTFT (ms)	P90 TTFT (ms)	P99 TTFT (ms)	Mean TPOT (ms)	Median TPOT (ms)	P90 TPOT (ms)	P99 TPOT (ms)	Mean ITL (ms)	Median ITL (ms)	P90 ITL (ms)	P99 ITL (ms)
3.5K	1.5K	10	2206.77	1941.75	3333.27	4469.28	49.99	47.43	62.69	77.29	79.15	75.37	100.85	195.94
32K	1K	10	2829.91	2296.63	4814.02	5387.56	55.92	54.84	66.12	85.09	84.45	79.02	107.03	493.89
32K	2K	10	3993.39	2253.39	11545.60	17175.45	53.26	52.29	73.42	86.09	84.75	79.35	108.49	204.45
64K	1K	10	4585.26	3990.01	7290.68	9458.98	57.04	56.69	71.08	87.38	88.63	83.12	110.19	532.27
64K	2K	10	5151.09	3893.12	12075.55	13530.88	55.02	54.60	69.26	77.27	85.50	81.61	110.05	197.75
128K	1K	10	432291.03	396537.29	669657.12	754730.19	66.60	67.95	77.09	82.33	79.87	76.96	108.68	180.88
128K	2K	10	529063.87	543963.16	756999.55	908553.75	68.33	68.52	84.34	117.99	89.84	77.39	107.99	232.54

4.3 P-DP4-TP8_D-DP8-TP4_Prefix cache 0%-MTP1（优化后）

输入	输出	并发	Mean TTFT (ms)	Median TTFT (ms)	P90 TTFT (ms)	P99 TTFT (ms)	Mean TPOT (ms)	Median TPOT (ms)	P90 TPOT (ms)	P99 TPOT (ms)	Mean ITL (ms)	Median ITL (ms)	P90 ITL (ms)	P99 ITL (ms)
2.05K	0.87K	20	2097.35	1647.73	3802.71	4478.28	48.87	47.49	56.34	71.38	72.39	72.31	75.70	120.10
2.05K	0.87K	23	2117.41	1624.71	3867.41	4623.33	48.80	45.53	63.88	73.04	73.24	73.15	77.51	130.41
3.5K	1.5K	10	1844.33	1562.59	3026.20	3784.20	49.25	43.92	72.15	73.73	73.63	73.52	76.38	136.27
3.5K	1.5K	11	1871.19	1537.71	3000.80	3801.20	50.28	45.81	71.84	72.16	73.38	73.43	75.64	135.63
20K	2K	1	4980.33	4936.05	5057.81	5085.21	72.99	72.95	73.94	74.16	76.01	74.97	108.14	154.98

结论：

Prefix cache 0，mtp=1

输入2.1K，输出0.9K，TPOT卡50 ms： 并发上限在23

输入3.5K，输出1.5K， TPOT卡50 ms：并发上限在10

输入20K，输出2K，TPOT卡50 ms：并发1，TPOT 超50ms，mtp=1和mtp=3均不满足

4.4 P-DP4-TP8_D-DP8-TP4_Prefix cache 90%-MTP1（优化后）

输入	输出	并发	Mean TTFT (ms)	Median TTFT (ms)	P90 TTFT (ms)	P99 TTFT (ms)	Mean TPOT (ms)	Median TPOT (ms)	P90 TPOT (ms)	P99 TPOT (ms)	Mean ITL (ms)	Median ITL (ms)	P90 ITL (ms)	P99 ITL (ms)
2.05K	0.87K	35	1812.11	1871.66	2596.90	2714.30	47.64	46.65	53.12	67.53	74.67	75.09	78.78	119.46
2.05K	0.87K	40	1794.86	1821.93	2356.59	2667.53	47.35	46.56	53.01	69.84	75.63	75.75	80.89	129.88
2.05K	0.87K	45	1990.75	2040.87	2894.37	3634.47	52.20	49.88	65.85	76.98	79.02	79.30	84.11	134.26
3.5K	1.5K	40	1811.01	1719.17	2844.60	2935.61	48.22	47.11	55.24	74.02	76.66	77.04	81.99	131.65
3.5K	1.5K	45	1875.39	1898.29	2431.32	2800.85	48.93	47.78	56.51	75.57	78.47	79.00	83.50	128.00
20K	2K	40	2403.09	1844.59	4355.55	5953.90	48.66	47.08	58.41	73.32	78.70	79.13	83.05	123.20
20K	2K	45	2525.54	1659.52	5097.89	7366.47	50.16	48.82	58.86	76.94	81.67	82.19	86.55	130.78
20K	2K	50	2469.29	1737.67	5214.61	6473.43	53.04	50.53	66.22	79.89	83.48	83.89	89.30	141.15

结论：

Prefix cache 90%，mtp=1

输入2.1K，输出0.9K，TPOT卡50 ms： 并发上限在40

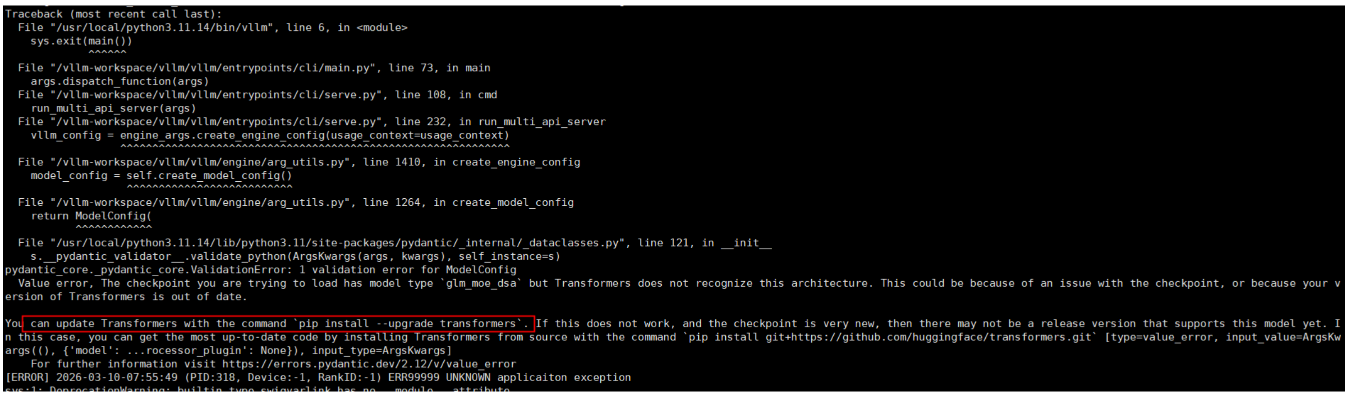
输入3.5K，输出1.5K， TPOT卡50 ms：并发上限在45

输入20K，输出2K，TPOT卡50 ms：并发上限在45

5. 问题

5.1 transformers 版本过低

问题现象：


```
Traceback (most recent call last):
  File "/usr/local/python3.11.14/bin/vllm", line 6, in <module>
    sys.exit(main())
             ^^^^^^
  File "/vllm-workspace/vllm/vllm/entrypoints/cli/main.py", line 73, in main
    args.dispatch_function(args)
  File "/vllm-workspace/vllm/vllm/entrypoints/cli/serve.py", line 108, in cmd
    run_multi_api_server(args)
  File "/vllm-workspace/vllm/vllm/entrypoints/cli/serve.py", line 232, in run_multi_api_server
    vllm_config = engine_args.create_engine_config(usage_context=usage_context)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/vllm-workspace/vllm/vllm/engine/arg_utils.py", line 1410, in create_engine_config
    model_config = self.create_model_config()
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/vllm-workspace/vllm/vllm/engine/arg_utils.py", line 1264, in create_model_config
    return ModelConfig(
           ^^^^^^^^^^^^
  File "/usr/local/python3.11.14/lib/python3.11/site-packages/pydantic/_internal/_dataclasses.py", line 121, in __init__
    s.__pydantic_validator__.validate_python(ArgsKwargs(args, kwargs), self_instance=s)
pydantic_core._pydantic_core.ValidationError: 1 validation error for ModelConfig
  Value error, The checkpoint you are trying to load has model type `glm_moe_dsa` but Transformers does not recognize this architecture. This could be because of an issue with the checkpoint, or because your v
ersion of Transformers is out of date.

You can update Transformers with the command `pip install --upgrade transformers`. If this does not work, and the checkpoint is very new, then there may not be a release version that supports this model yet. I
n this case, you can get the most up-to-date code by installing Transformers from source with the command `pip install git+https://github.com/huggingface/transformers.git` [type=value_error, input_value=ArgsKw
args((), {'model': ...rocessor_plugin': None}), input_type=ArgsKwargs]
    For further information visit https://errors.pydantic.dev/2.12/v/value_error
[ERROR] 2026-03-10-07:55:49 (PID:318, Device:-1, RankID:-1) ERR99999 UNKNOWN applicaiton exception
```

解决方案：升级transformer版本

```
pip install transformers==5.2.0 --no-deps --force-reinstall
pip install huggingface_hub==1.5.0 --no-deps --force-reinstall
```

5.2 加了rot.safetensors，报错 KeyError: 'rot'

问题现象：

```
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783] WorkerProc failed to start.
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783] Traceback (most recent call last):
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm/vllm/v1/executor/multiproc_executor.py", line 754, in worker_main
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     worker = WorkerProc(*args, **kwargs)
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm/vllm/v1/executor/multiproc_executor.py", line 580, in __init__
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     self.worker.load_model()
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm-ascend/vllm_ascend/worker/worker.py", line 431, in load_model
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     self.model_runner.load_model()
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm-ascend/vllm_ascend/worker/model_runner_v1.py", line 2490, in load_m
odel
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     self.model = get_model(vllm_config=self.vllm_config)
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm/vllm/model_executor/model_loader/__init__.py", line 135, in get_mod
el
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     return loader.load_model(
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]            ^^^^^^^^^^^^^^^^^^
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm/vllm/model_executor/model_loader/base_loader.py", line 62, in load_
model
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     self.load_weights(model, model_config)
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm/vllm/model_executor/model_loader/default_loader.py", line 290, in l
oad_weights
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     loaded_weights = model.load_weights(self.get_all_weights(model_config, model))
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]   File "/vllm-workspace/vllm/vllm/model_executor/models/deepseek_v2.py", line 1483, in load_weig
hts
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]     param = params_dict[name]
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783]             ~~~~~~~~~~~^^^^^^
(Worker_DP0_TP0_EP0 pid=1942) ERROR 03-12 17:53:35 [multiproc_executor.py:783] KeyError: 'rot'
```

解决方案：

rot的权重要放在其他目录，不能和w8a8的放一起

5.3 部署上下文200K，报超出模型最大支持长度

问题现象：

```
Value error, User-specified max_model_len (204800) is greater than the derived max_model_len (max_position_embeddings=202752.0 or model_max_length=None in model's config.json). To allow overriding this maximum, set the env var VLLM_ALLOW_LONG_MAX_MODEL_LEN=1. VLLM_ALLOW_LONG_MAX_MODEL_LEN must be used with extreme caution. If the model uses relative position encoding (RoPE), positions exceeding derived_max_model_len lead to nan. If the model uses absolute position encoding, positions exceeding derived_max_model_len will cause a CUDA array out-of-bounds error. [type=value_error, input_value=ArgsKwargs((), {'model': ...rocessor_plugin': None}), input_type=ArgsKwargs]
```

问题解决：

将`--max-model-len`参数调低，官方宣称最大支持200K上下文，实测最大只能到198K

5.4 PP并行策略报错

问题现象：

```
NotImplementedError: Pipeline parallelism is not supported for this model. Supported models implement the `SupportsPP` interface.
```

问题解决：

模型不支持PP并行策略，改用DP+TP并行策略